Course Syllabus
Computer Organization and Assembly Language Programming

1. COURSE INFORMATION

Course & Section Number:	CIST 039 – 32128, 4 Semester Units
Semester & Year:	Spring 2026
Lecture (No Lab):	Tuesday & Thursday, 8:00 AM – 10:05 AM, Che 131 and HyFlex
Final Date and Time:	Tuesday, May 19h, 7:30AM - 9:30, Che 131
Drop Policy:	Without 'W': Feb 8th; With 'W': April 24th; Pass/Fail: May 14th
Required Prerequisite:	Any Programming Class, (Required)
Recommended Prerequisite:	Discrete Math and Data Structures (any programming language)

2. INSTRUCTOR INFORMATION

Name:	Howard Miller
Student (AKA Office) Hours:	Tue & Thur 10:10 AM-11:10 AM, Che 105A or on Zoom And by Arrangement
Cell Phone / Text:	408-802-4034 (Text 7:30 AM - 10 PM ONLY!)
E-mail:	howard.miller@wvm.edu

3. INSTRUCTION FORMT

Synchronous Live Feed:	In-person Che 131, AND Zoom Conferencing, Synchronous
Course Management:	Canvas (WVC Login, Valid & Active eMail account required)
Course Collaboration:	Piazza.com and Discord.com (You must create accounts)

4. REQUIRED COURSE MATERIALS

Title:	Computer Organization and Design ARM Addition
Author:	David A. Patterson & John L. Hennessy
Publisher:	Morgan Kaufmann 978-0128017333
Assembly Programming Text:	http://personal.utdallas.edu/~pervin/RPiA/RPiA.pdf
Hardware:	Raspberry Pi3B+ / 4 / 5 or Pi400/500 2GiB+ w/Case & Power Supply
Storage:	USB Flash Drive: 64MB or greater
Display / Computer:	HDMI Monitor, Keyboard and Mouse - OR - Mac or PC
Required Student Accounts:	Canvas: With valid & active eMail address Piazza: https://piazza.com/westvalley/spring 2026/cist039 Piazza Passcode Code given in lecture/Canvas is required
Optional Student Account:	Discord: https://discord.gg/rd6hNTdgRW

*** Recommended Supplemental Materials and Resources**

Assembly Program on Raspberry Pi / ARM
- http://bob.cs.sonoma.edu/IntroCompOrg-RPi/intro-co-rpi.html

ARM References
- http://developer.arm.com/documentation/ddi0500/j
- http://developer.arm.com/documentation/100095/0003
- http://www.keil.com/support/man/docs/armasm/armasm_dom1361289850039.htm

C Programming
- https://www.learn-c.org

5. COURSE DESCRIPTION

This course introduces the fundamental concepts of computer organization and assembly language programming. It studies the basic Instruction Set Architecture and hardware of computer processors. It explores how computer systems execute programs and manipulate data, working from a high level programming language down to the hardware. Topics covered include: The C programming language briefly, Assembly Language programming, machine-level code, data representation (integer, floating-point, character and string representations), computer arithmetic: Integer and floating point, memory organization, hierarchy & management, performance evaluation and optimization, power constraints, reliability through redundancy, program parallelism in both software and hardware, I/O interfacing and interrupts. For the lab / programming work, this course will study and program on the most popular computer architecture, the ARM family of processors.

This course focuses the Architecture and Organization Knowledge Area prescribed by the Association of Computer Machinery Computer Science Curricula 2023 including AR-2023 with supplemental topics from the Knowledge Areas: Operating Systems, Parallel and Distributed Computing, and System Fundamental. Learning is reinforced with written homework and programming exercises for the ARM architecture in the Linux environment.

6. STUDENT LEARNING OUTCOMES

Official Student Learning Outcomes:

- Create a program using computation and operations using the following Assembly programming fundamentals: bit manipulation, arrays, recursion, advanced data types, macros, and pseudo operations.
- Debug, test and run a program involving Assembly programming using numbers, strings and memory management.

Upon successful completion of this course, students will be able to:

- Understand computer architecture and the von Neumann and Harvard Architectures
    - History of Computer Architecture
    - Control unit; Memory; Instruction fetch, decode, and execution
    - Data Encoding (Bits, Bytes, Words, Integers, IEEE 754, ASCII, Unicode)
    - Instruction sets (ISA) and Instruction formats and encoding
    - Addressing modes
    - Assembly/Machine language programming using load/store, arithmetic, logic, branches, call/return and stack instructions.
    - Runtime memory layout: Static Data vs. Stack and Static Text vs. Code section
    - Show how fundamental high-level programming constructs are implemented at the assembly/machine-language level.
    - Subroutine call and return mechanisms.
    - Exception and Interrupt handling mechanisms.
    - Usage of the AAPCS Application Binary Interface (ABI), Application Programming Interfaces (APIs) and other Call Conventions
    - The evolution of hardware/software techniques and application needs
    - Describe instruction level parallelism and hazards, and how they are managed in

typical processor pipelines.

- Understand key concepts of Memory System Organization and Architecture
    - Storage systems and their technology
    - Memory hierarchy: importance of temporal and spatial locality for Caching
    - Main Memory organization and operations
    - Virtual Memory, Page Tables and TLB Implementation
- Machine Level Representation of Data
    - Convert between Integer decimal, binary, and hexadecimal notations.
    - Work with two's complement integers
    - Decode/Encode IEEE 754 Floating Point formats
    - Characteristics and limitations of Floating Point arithmetic
    - Work with (encode / decode) ASCII and UTF character encodings
    - Understand the representation of records and arrays
- Digital Logic and Digital Systems
    - To be familiar with the architectural components of a computer system: CPU (registers, ALU), memory, buses.
    - Physical constraints of space, energy/power and defects
- Parallel Algorithms, Analysis, and Programming
    - Speed up measurement, computation and comparison
    - Amdahl's law and its application
- Computer Reliability and Availability
    - Parity, Error Detection and Error Correction. Hamming Codes.
    - Redundant Array of Inexpensive Disks.
    - Virus and Ransomware Attacks
    - Data recovery: 3-2-1 Back-up, Use of Snapshots and Copy-On-Write filesystems

Assignments: On-line quizzes and homework problem sets based on the course textbook will be regularly assigned. Writing programs in ARM assembly language is an integral part of this course and number will be assigned. Programs are graded on accuracy, style, readability and maintainability. Assignments must be handed-in on time. Homework and Programs not meeting the minimum standard of correctness, completeness and coding style will need to be corrected. You must achieve at least 75% aggregate score on the total of both all programming, and all homework assignments to receive credit for this course. **No late homework will be accepted without Prior Approval of the Instructor.**

7. GRADING

A 90 – 100%
B 80 – <90%
C 70 – <80%
D 60 – <70%
F <60 (Or Less than 75% of programs OR less than 75% of Homework)

I. 25% Programs (75% or more Required for a passing grade)
II. 30% Homework (75% or more Required for a passing grade)
III. 5% Pre-work, (on Canvas), Piazza Posts plus In-class pop-Quizzes
IV. 20% Midterms (2, 10% each)
V. 20% Final exam

8. TENTATIVE COURSE OUTLINE

Week -	Topics (Chapter)
Week 1 -	Chapter 1 Computer Abstraction and Technology
Week 2 -	ARM Programming: Intro to Raspberry Pi, Hello Program
Week 3 -	ARM Programming: AAPCS, Square Program
Week 4 -	Chapter 2 Instructions: Language of the Computer
Week 5 -	Chapter 2 Instructions: Hex to Binary Decode Program
Week 6 -	Chapter 2 Instructions: Encoding / Decoding Program
Week 7 -	Midterm Review and Midterm 1
Week 8 -	Chapter 3: Computer Arithmetic with Floating Point
Week 9 -	ARM Programming: Floating Point Decode Program
Week 10 -	Chapter 4: The Processor, The Data Path and Pipeline
	Chapter 4: The Processor, Hazards and their Control
Week 11 -	Chapter 4: Interrupts, Branch Prediction / Midterm Review
Week 12 -	Midterm 2 ; Functions and C-strings Program
Week 13 -	Chapter 5 Exploiting the Memory Hierarchy: Caches
Week 14 -	Chapter 5: VM and Dependability (ECC and RAIDs)
Week 15 -	Chapter 6: Parallel Processors overview; Review for Final
Week 16 -	Final Exam (Tuesday 7:30 AM – 9:30 AM)
	**Subject to Change / adjustment

9. CLASS ATTENDANCE

Students are expected to attend Synchronously All sessions of each class. Classes include programming time in selected lectures. Instructors may drop students from the class if they fail to attend the first class meeting, or when accumulated unexcused hours of absences exceed ten percent of the total number of hours the class meets during the semester. Moreover, an instructor may drop from the class any student who fails to attend at least one class session during the first three weeks of instruction.

10. PREFERRED NAME AND PRONOUN

If you'd like to be known by a name different from the name on the roll sheet or have a preferred pronoun, please indicate that in you on-screen zoom name. You can change your name in Canvas by contacting Max Gault <max.gault@westvalley.edu>. You can identify your preferred pronoun in Canvas. https://community.canvaslms.com/t5/Student-Guide/How-do-I-select-personal-pronouns-in-my-user-account-as-a/ta-p/456

11. ONLINE APPLICATIONS

Canvas is a course management system adopted by the WVMCCD for all classes. When you log into the system, you will see a listing of classes that you are taking. wvm.instructure.com/login/canvas

Zoom Video Conferencing has been adopted by the California Community Colleges for on-line classes. This can be accessed via their application or a compatible web browser. Video and Audio will be used in this course. No account is necessary. The course URL will be publishing on Canvas.

Piazza is learning management system designed for Computer Science students to share program hints and tips (not answers). We will use this system for this course. http://piazza.com/westvalley/cist039

12. CHEATING POLICY

Dishonesty includes but is not limited to in-class cheating, out-of-class cheating, plagiarism, use of AI, knowingly assisting another student in cheating or plagiarism, or knowingly furnishing false information to college staff, faculty, administrators or other officials. Following are definitions of in-class cheating, out-of-class cheating, plagiarism, and furnishing false information. These are not all-inclusive and the list itself is not meant to limit definition of cheating to just those mentioned.

a. In-class cheating: during an examination or on any work for which the student will receive a grade or points, unauthorized looking at or procuring information from any unauthorized sources, or any other student's work.
b. Out-of-class cheating: unauthorized acquisition, reading or knowledge of test questions prior to the testing date and time; changing any portion of a returned graded test or report and resubmitting as original work to be re-graded; or presenting the work of another as one's own for a grade or points; using test notes not created by you or directly provided by the instructor.
c. Plagiarism: unauthorized use of expression of ideas from AI generated, or published work(s) or unpublished work(s) or AI generated as a student's own work for a grade in a class. This also includes the violation of copyright laws, including copying of software packages.
d. Furnishing false information: forgery, falsification, alteration or misuse of college documents, records, or identification in class or in laboratory situations.

Those found cheating will have their grade reduced, or will be removed from the course. There is no Statute of Limitations for cheating; If found cheating, your grade can be reduced at any time.

13. CODE OF STUDENT CONDUCT

The District shall enforce a student code of conduct the purpose of which is to promote and maintain orderly conduct of a responsible student body in a manner compatible with the District and College function as an educational institution.
https://www.westvalley.edu/policies/student-conduct.html

14. DISABILITY STATEMENT

ADA Statement: The American with Disabilities Act (ADA) is a federal anti-discrimination statute that provides comprehensive civil rights protection for persons with disabilities. Among other things, this legislation required that all students with disabilities be guaranteed a learning environment that provides for reasonable accommodation of their disabilities. Disability and Educational Support Program (DESP) supports students with learning and other disabilities to achieve their educational goals through comprehensive support and services including specialized counseling, learning disability services, alternate media and adapted physical education; call 408-741-2010; TTY: 408-741-2658.

15. NON-DISCRIMINATION STATEMENT

The District, and each individual who represents the District, shall provide access to its services, classes, and programs without regard to national origin, religion, age, gender, gender identity, gender expression, sex, race or ethnicity, color, medical condition, genetic information, ancestry, sexual orientation, marital status, physical or mental disability, pregnancy, or military and veteran status, or because he/she is perceived to have one or more of the foregoing characteristics, or based on association with a person or group with one or more of these actual or perceived characteristics.

16. SAFETY/EMERGENCY

It is the student's responsibility to know the evacuation procedures, evacuation route, and assembly area for this classroom. In case of an emergency, you are to follow the directions of your instructor. When directed to evacuate the classroom, *be sure to take all of your belongings when you leave* and remain with your class in the assembly area until you receive further directions.

Be prepared! Please review the college's information:
https://wvm.edu/services/police/prepare/Pages/default.aspx

WVM-Alert **- Emergency Notification**

WVM-Alert will text, email and call you to alert you to campus emergency situations. Sign in to https://www.wvm.edu/services/police/alert/Pages/default.aspx and give us your contact information ASAP! If you don't sign up, you won't be notified!

17. FEES AND TUITION

Payment Plan Options: West Valley College is committed to partnering with **Nelnet** to make your educational dreams possible. You have the opportunity to select a payment plan to make your enrollment expenses more affordable. Additional information can be found on the WVC Portal. Note: Students are responsible for paying **all enrollment fees in full**.

Student Responsibility: Once a student has registered for classes, the student is considered officially enrolled. It is the responsibility of the student to drop or withdraw completely otherwise the student will be accountable for all enrollment and related fees for the classes. Failure to attend class meetings does not constitute withdrawal from the class and does NOT exempt a student from the obligation to pay all enrollment and related fees.
http://westvalley.edu/admissions/fee-schedule.html

Instructional Material Fee: It is the policy of the West Valley-Mission Community College District that the Governing Board may require students to provide instructional and other materials required for credit and non-credit courses, provided that such materials are of continual value to a student outside of the classroom setting and provided that such materials are not solely or exclusively available from the District.

18. COUNSELING SERVICES

We encourage students to complete orientation and educational planning through the Counseling Services office in order to maximize their success. Data shows that students who have complete their educational planning and orientations before registering for classes have a higher success and completion rate than those who do not. For appointments or drop in information: call 408-741-2009 or just walk into the Counseling Building during open hours.

19. STUDENT RIGHT TO KNOW

Please refer to the following links for information on the federally mandated requirements regarding a student's right to be informed about college data on graduation and completion rates and other information: http://www.westvalley.edu/policies/index.html
https://www.westvalley.edu/policies/heoa.html

20. TUTORING INFORMATION

Tutorial Services: Tutoring is available, without charge, to West Valley College Students. The service is provided by trained, qualified, students and professionals who have been recommended by faculty. Students are tutored in a drop-in, group, or individual environment depending on subject of interest. Most subjects are available. Students must be currently enrolled at West Valley College in the subject for which they request help. Study groups are welcomed and encouraged. To sign up for tutoring, come to the Tutorial Services office, or visit:
http://westvalley.edu/services/academic-success/tutorial/index.html

Writing Center: The Writing Center is an open computer lab where students can get help with writing assignments for any class at WVC. Peer tutors and English faculty members are available to help you succeed. Open lab hours and one-on-one tutoring are available. The Writing Center provides: professional faculty members, drop-in individual tutoring, and a lab with computers, writing and reading resources, help with basic grammar, mechanics, essay organization, and much more.
http://westvalley.edu/services/academic-success/tutorial/index.html

STEM Resource Center (STEM RC): The STEM Resource Center (STEM RC), located in SM 5, supports West Valley College Students success in mathematic courses. Free tutoring is offered in all levels of mathematics. The MRC is staffed by math faculty and peer tutors. For more information please visit our website.
https://www.westvalley.edu/academics/mathematics/math-resource-center.html

21. THE FAMILY EDUCATIONAL RIGHTS AND PRIVACY ACT (FERPA)

(20 U.S.C. § 1232g; 34 CFR Part 99) is a Federal law that protects the privacy of student education records. The law applies to all schools that receive funds under an applicable program of the U.S. Department of Education. For additional information, you may call 1-800-USA-LEARN (1-800-872-5327) (voice). Individuals who use TDD may call 1-800-437-0833.

22. SEXUAL HARASSMENT AND SEX DISCRIMINATION

In accordance with Title VII Section 1604, and Title IX of the 1972 Education Amendments, it is the policy of the West Valley-Mission Community College District to provide an educational, employment and business environment free of unwelcomed sexual advances, requests for sexual favors, and other verbal or physical conduct or communications constituting sexual harassment and/or sex discrimination as defined and otherwise prohibited by Federal and State law. Complaints by students or employees should be directed to Associate Vice Chancellor of Human Resources. The telephone number is (408) 741-2131.

Campus SAVE (Sexual Violence Elimination) Act: refer to the following link for information on the federally mandated protections for victims of sexual violence. https://www.westvalley.edu/campus-safety/

23. Illnesses (Covid and Other)

Please be respectful of others and do NOT come to campus when you are ill! If you feel bad, you can attend via Zoom, just let me know via text that is your plan for the day.

24. STUDENT HEALTH SERVICES

Clinical visits with a nurse; personal counseling for stress, anxiety, depression and relationship concerns; low cost prescriptions, immunizations; and support for victims of domestic violence, alcohol and substance abuse and sexual assault are all available through our excellent Student Health Services; call 408-741-2027; urgent contact line: 408-741-4000. https://www.wvm.edu/services/health/

25. WEST VALLEY COLLEGE IS A NO-SMOKING, NO-VAPING AND DRUG-FREE CAMPUS

West Valley College is a non-smoking, non-vaping environment. This is reflected in District policy: Smoking, the use of smokeless tobacco and the use of e-cigarette devices are prohibited in all areas of the Mission and West Valley campuses except in parking lot areas that are at least twenty-five (25) feet away from buildings and pathways. The West Valley-Mission Community College District policy prohibits "the unlawful use, distribution, sale, or possession of alcohol, narcotics, dangerous or illegal drugs, or other controlled substances, as defined in California statues, on District property or at any function sponsored by the District or colleges."

26. STUDENT NEEDS

Food: If you need help affording food while attending West Valley College, you are not alone, and West Valley College can help. A mobile food pantry provides free food on campus weekly.

Housing: West Valley has resources to help you deal with housing instability or homelessness. West Valley also has on-campus showers Monday -Thursday.

Finances: If you are having difficulty paying for your classes, there is help.
https://www.westvalley.edu/services/student-needs/

27. GRIEVANCE PROCESS

The grievance process is a formalized process to ensure the timely resolution of conflict at the lowest possible level. The first step is the informal resolution stage, which involves the student who has a complaint and the staff member or specific group who is the other party in the grievance. The student must notify the staff person or representative of a group that she/he wishes to make an appointment for an informal meeting to review an action within ten (10) days of its occurrence. In the absence of the instructor or staff person and after a good faith effort to make contact, the grievant may directly contact the department chair. Additional information is available from the Vice President of Student Services.

28. POLICY FOR COURSE REPETITION

Title 5 code 55040: District Policy for Course Repetition. A student may repeat any course in which a substandard final grade (D F, NP, or W) was earned. A course may be repeated only once under this policy for a total of two attempts. A student wishing to repeat a course for a 3rd attempt will be required to submit a Student Petition Form.

29. LAND ACKNOWLEDGEMENT

We acknowledge that West Valley College sits on the land of the Ohlone and the Muwekma people. For thousands of years, these natives occupied this land and used this beautiful location as their home. Let us give an enormous debt of gratitude to the Ohlone and Muwekma tribes. With this Acknowledgment, we remember that the Ohlone and Muwekma people are still connected to this region.